



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start!
- Let's Explore!
- Let's Practice!
- Extension Activity
- Target Words
- Learning Points



Let's Start!

Connecting Servers and Sharing Data!



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognizes and implements the different types of architecture styles that can be included in the IT solution.

Can Do Statements:

- I can talk about rule-based architecture style.
- I can describe peer to peer architecture style.
- I can give information about publish-subscribe architecture style.

Ask your partner:

- What applications or programs to share files do you know?
- What are the advantages of allowing systems to provide answers to problems found by users?





Let's Explore!

Leonard teaches Sonia about architecture styles!



Instructions: Read the text below.

Sonia: What is your favorite app to listen to music?

Leonard: I know Spotify® is considered more popular but I prefer Napster®. Anyway they follow similar peer-to-peer principles.

Sonia: What do you mean?

Leonard: Let me describe to you what peer-to-peer is and what to take into account. It's a network where all peers are designed to be suppliers and consumers of services. Peers communicate by nodes and the type of connection depends on the network, if it's structured, the system organizes the protocol to ensure that files are found easily. If the network is unstructured, there are random connections among peers. These styles are used in public-sharing apps such as Napster®. Spotify® uses a hybrid model, though.

Sonia: I thought that the architecture style they used was called publish-subscribe.

Leonard: Not really. Publish-subscribe pattern is about data sent by publishers and obtained by subscribers. Subscribers decide what to view since the publishers don't send the messages directly and they can filter the information by title or content. Unfortunately, publishers don't know exactly what to post since they need to know the subscribers' interests. One big advantage of this style is the scalability that it can reach. It is commonly used for databases or mailing lists.

Sonia: I see, that's the same style that expert systems uses!

Leonard: Actually, no! They use a rule-based system. You tell the system what to interpret and it analyzes the axioms in order to give logic answers to problems. This system uses a semantic reasoner that matches the information, establishes a conflict-resolution process, and acts accordingly. Basically, the reasoner infers information and takes action based on the input that it has. Rule-based system are used as compilers to interpret computer programs.

Sonia: I forgot you were an expert in architecture styles!



**Let's Practice!**

A rule-based architecture style has these characteristics.

Instructions: Read the sentences. write the missing words. Use: is related, obtained by, taken into account, filtered by, commonly used.

1. Publish-subscribe pattern is about data sent by publishers and _____ subscribers.
2. The information is _____ the publishers in a title or content manner.
3. Each post _____ to the subscribers' interest, for this reason, sometimes publishers don't know what to show.
4. The scalability that it can reach is _____ as a big advantage that this style has.
5. This style is _____ for databases or mailing lists.

**Let's Practice!**

She didn't know what to do with this architecture style!

Instructions: Cross out the incorrect word and write in the correct word. Practice reading the sentences aloud and check your pronunciation with your instructor.

1. I'm wondering if she knows _____ the peer-to-peer architecture style. (how to work / what to work).
2. She doesn't know _____ account to use the peer-to-peer architecture style. (what to taking into. / what to take into)
3. She didn't know _____ kind of architecture style in the system. (where to find this / where to found this)
4. I wonder if she knows _____ the peer-to-peer architecture style. (how to uses / how to use)
5. She doesn't know _____ the information in this architecture style. (who to ensure / how to ensure)





Extension Activity

What could be the best style for you?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about rule-based, peer to peer and publish-subscribe architecture style.
- Take notes about the information of each architecture style.
- Make a description with your own words about each architecture style.
- Make a brochure with your description to promote each architecture style.
- Share your brochure with your partners.

Self-Assessment

Yes

No

- I look for information about rule-based, peer to peer and publish-subscribe architecture style.
- I make a description of each architecture style.
- I share it with my partners.

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Target Words

- **axiom** (noun): statements that a system considers true by logic
e.g. You tell the system what to interpret and it analyzes the axioms in order to give logic answers to problems.
- **compiler** (noun): program that converts code language into programming language
e.g. Rule-based system are used as compilers to interpret computer programs.
- **peer** (noun): something or someone with the same characteristics
e.g. Peers communicate by nodes and the type of connection depends on the network.
- **infer** (verb): to interpret a message by using logics
e.g. Basically the reasoner infers information and takes action based on the input that it has.
- **semantic reasoner** (noun): software that infers logical consequences from facts
e.g. This system uses a semantic reasoner that matches the information, establishes a conflict-resolution process, and acts accordingly.
- **subscriber** (noun): receiver of messages provided by a publisher
e.g. Subscribers decide what to view since the publishers don't send the messages directly.



Learning Points

What to + verb.

You use **what to + verb** to give emphasis to an object or thing before the verb.

What refers to an object or a thing, it helps us provide a content to a sentence depending on the action.

Let's review these sentences:

*Publishers only care about **what they post**.*

*Publishers only care about **what to post**.*

In the previous sentences we can see how the verb **post** gives meaning to **what**. Now, instead of repeating the subject we can give the same meaning to the sentence by using **what to**.

Examples

- *Subscribers decide what to view since the publishers don't send the messages directly and they can filter the information by title or content.*
- *Unfortunately publishers don't know exactly what to post since they need to know the subscribers' interests.*
- *You tell the system what to interpret and it analyzes the axioms in order to give logic answers to problems.*

Rule-based, publish-subscribe and peer to peer architecture styles.

These architecture styles are commonly used for problem solving, sharing data, and obtaining data.

Rule-based has to do with ways to manage data, usually manipulated by artificial intelligence applications.

Publish-subscribe refers to senders of messages called publishers and receivers called subscribers.

Peer-to-peer is a method to distribute services between peers that work as consumers and servers at the same time.

In the following chart, we will see some characteristics of each type.





Learning Points

Examples

- *This system uses a semantic reasoner that matches the information, establishes a conflict-resolution process, and acts accordingly.*
- *Publish-subscribe pattern is about data sent by publishers and obtained by subscribers.*
- *It's a network where all peers are designed to be suppliers and consumers of services.*